

# Radhika Sishodiya

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## Technologies

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**Languages:** Java , Python ,SQL

**Backend:** Spring Boot, Spring Security, REST APIs

**Frontend:** HTML , CSS

**Databases:** PostgreSQL

**Tools:** Git, GitHub, MAVEN, DOCKER, POSTMAN,VERCEL

## Open Source Contribution

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### Karate-Championship-Management-App [🔗](#)

- Contributed to Karate-Championship-Management-App, a Java 21 / Spring Boot backend project for managing competitive karate tournaments.

## Projects

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### StudySync – AI-Native Collaborative Learning Platform — Java, Spring Boot, Spring AI, WebSocket, PostgreSQL, OAuth2 [🔗](#)

- Built an AI-assisted collaborative learning platform with a React/TypeScript frontend and Spring Boot backend, enabling real-time study rooms and on-demand AI assistance.
- Built real-time collaboration, authentication, and AI-assisted learning features.
- Integrated **Spring AI** with **GPT-4o-mini** to provide four assistance modes: explain, quiz, summarize, and interview.
- Centralized authorization in a fail-closed access-control service, validating permissions for every WebSocket subscription and REST request before allowing room-level actions.
- Implemented stateless **JWT authentication** with HMAC-signed access tokens and DB-backed refresh tokens, alongside **Google and GitHub OAuth2** login.
- Implemented real-time messaging using **STOMP/SockJS** with room-scoped topics and a personal notification queue, using a persist-then-broadcast pattern to ensure messages are stored before delivery.

### PayFlow – Digital Wallet & Transaction Service — Java, Spring Boot, Spring Security, PostgreSQL, Docker [🔗](#)

- Built a digital wallet backend for concurrent money transfers, addressing double-spend and inconsistent-balance risks.
- Implemented **SERIALIZABLE** transaction isolation with pessimistic row locking and consistent **UUID-based lock ordering** to prevent deadlocks during concurrent transfers.
- Added a database-level **unique constraint on idempotency keys** to prevent duplicate transactions caused by client retries.
- Developed unit tests for insufficient balance, self-transfer, duplicate idempotency key, and invalid receiver scenarios, plus **Testcontainers integration tests** executed through GitHub Actions CI on every push and pull request.

## Education

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**Dr. A.P.J. Abdul Kalam Technical University, ABESEC**  
Computer Science and Engineering, CGPA: 8.1

*Expected July 2027*

## Certificates

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- Spring Boot 0 to 100 Cohort 4.0 (AI + DevOps)
- AWS Academy Graduate- foundations in cloud computing concepts and core AWS services.